

The root class search user study

Thank you for participating in the user study.

Introduction

The Dataspecer tool enables users to model a tree-like data structure based on an input ontology. The data structure can be used to generate various software artifacts. The ontology is a set of classes and properties. Properties are divided on attributes (a value is a literal) and associations (a value is a class). The data structure creation starts with a root class selection. Then the user models the data structure by adding properties of the root class. If the properties are associations, the user continues recursively on their endpoint classes.

Your Task

You are assigned four descriptions of a root class listed as part of the questionnaire. For each description, you are tasked to search for an appropriate root class of the data structure in the Wikidata ontology using the Dataspecer tool.

The descriptions contain only information about the root classes and the properties the classes should have. **Your task is to only find the root class. Do not model the entire data structure.**

Step-By-Step instructions

1. Open the search dialogue in the Dataspecer tool:
 1. Open [the Dataspecer tool](#).
 2. Click the "Create specification" button, enter any label you want, and click "Create".
 3. In the "Vocabulary sources", **select Wikidata and click on the button "save"**.
 4. Proceed by clicking the large rectangle "Create data structure" button at the top of the page.
 5. Click the "Set root element".
 - 6. In the top side of the panel, select method B.**
 7. You have successfully opened the search dialogue you will be using to complete the tasks.
2. Follow the questionnaire, each task is in a different section.

Between tasks, you can simply close and reopen the dialog, or you can just simply continue without closing the dialog. If you reopen the dialog do not forget to select the **method B** again.

Search Dialog Tips

- By clicking on a class in the list of results, the class will be selected as a root

class.

- **You can view the class properties by opening a detail of the class, by clicking on the "(i)" icon and navigating to a "Wikidata" panel.**
 - You can input keywords, a short description, full sentences or any combination of the mentioned.
 - Use the properties filter that enables you to select properties the class must have.
 - Use the slider option to boost classes with many instances. The slider denotes the weight of the boost.
-

* Indicates required question

1. Characterize yourself *

Tick all that apply.

- ☐ A student
- ☐ A teacher
- ☐ A modelling expert
- ☐ A programmer
- ☐ A manager

Search Task 1

Find the root of a data structure for exchanging data about defensive military buildings designed to protect people and materials from enemy attacks. The buildings were extensively used during World War I-II and usually contained many floors below ground. The data will be used to display information panels in a war museum. The root needs to represent information about conflicts and significant battles the building participated in. Information about the building's schema, capacity, and time it entered and retired service must also be included. Lastly, the panel must state the current state of conservation.

When searching, **try both with and without** the property filter and boosting. Give yourself **7 minutes at most** before continuing. Do not forget to select the **method B** again.

2. I understood the root description clearly. *

Mark only one oval.

1	2	3	4	5		
Fully	☐	☐	☐	☐	☐	Fully agree

3. I have managed to find a reasonable root class. *

Mark only one oval.

1	2	3	4	5		
Fully	☐	☐	☐	☐	☐	Fully agree

4. Enter the ID of the found root class. If you did not find a class, enter "no". *

The ID is located next to the class name, e.g. for a class "human (Q5)", the ID is the Q5.

Search Task 2

Find the root class of a data structure for exchanging data about schools/universities that enable enrollment of both male and female students. The data will form an online brochure for educational institutes' fairs. The root class needs to be associated with famous people who were educated at the institute, the number of current students, a name, and methods used for teaching. Topics/subjects included in the curriculum must also be present.

When searching, **try both with and without** the property filter and boosting. Give yourself **7 minutes at most** before continuing. Do not forget to select the **method B** again.

5. I understood the root description clearly. *

Mark only one oval.

1 2 3 4 5

Fully ☐ ☐ ☐ ☐ ☐ Fully agree

6. I have managed to find a reasonable root class. *

Mark only one oval.

	1	2	3	4	5	
Fully	☐	☐	☐	☐	☐	Fully agree

7. Enter the ID of the found root class. If you did not find a class, enter "no". *

The ID is located next to the class name, e.g. for a class "human (Q5)", the ID is the Q5.

Search Task 3

Find the root class of a data structure for exchanging data about general chemical substances and entities. The class must be general enough to encompass all objects of interest in chemistry. The most important properties are the description of the chemical formula accompanied by a chemical structure and the weight of the chemical object. Additionally, the class must provide properties for the effects of the objects, an international chemical identifier (InChI), and a list of other objects it interacts with.

When searching, **try both with and without** the property filter and boosting. Give yourself **7 minutes at most** before continuing. Do not forget to select the **method B** again.

8. I understood the root description clearly. *

Mark only one oval.

	1	2	3	4	5	
Fully	☐	☐	☐	☐	☐	Fully agree

9. I have managed to find a reasonable root class. *

Mark only one oval.

	1	2	3	4	5	
Fully	☐	☐	☐	☐	☐	Fully agree

10. Enter the ID of the found root class. If you did not find a class, enter "no". *

The ID is located next to the class name, e.g. for a class "human (Q5)", the ID is the Q5.

Search Task 4

Find the root class of a data structure for exchanging data about administrative territorial entities that provide municipal services in Luxembourg. The data will be provided for Luxembourg's Open Data Catalogue. The root class must provide a name and statistical information about the entities. The class should represent the population, the unemployment rate, the number of households, the average income, and the incidence of poverty. Furthermore, an associated electoral and cadastral district must be present.

When searching, **try both with and without** the property filter and boosting. Give yourself **7 minutes at most** before continuing. Do not forget to select the **method B** again.

11. I understood the root description clearly *

Mark only one oval.

1	2	3	4	5		
Fully	☐	☐	☐	☐	☐	Fully agree

12. I have managed to find a reasonable root class *

Mark only one oval.

1	2	3	4	5		
Fully	☐	☐	☐	☐	☐	Fully agree

13. Enter the ID of the found root class. If you did not find a class, enter "no". *

The ID is located next to the class name, e.g. for a class "human (Q5)", the ID is the Q5.

The questions about the search dialog

14. The search engine returned relevant results to the provided queries. *

Mark only one oval.

1 2 3 4 5

Fully ☐ ☐ ☐ ☐ ☐ Fully agree

15. I found the most relevant results at the top of the result's list. *

Mark only one oval.

1 2 3 4 5

Fully ☐ ☐ ☐ ☐ ☐ Fully agree

16. I consider the property filter useful. *

Mark only one oval.

1 2 3 4 5

Fully ☐ ☐ ☐ ☐ ☐ Fully agree

17. I consider the boosting of widely used classes useful. *

Mark only one oval.

	1	2	3	4	5	
Fully	☐	☐	☐	☐	☐	Fully agree

18. The information provided in the dialog was sufficient for me to select an appropriate root class. *

Mark only one oval.

	1	2	3	4	5	
Fully	☐	☐	☐	☐	☐	Fully agree

19. The search engine was sufficiently responsive. *

Mark only one oval.

	1	2	3	4	5	
Fully	☐	☐	☐	☐	☐	Fully agree

20. I found the search engine intuitive to use. *

Mark only one oval.

	1	2	3	4	5	
Fully	☐	☐	☐	☐	☐	Fully agree

21. The search engine enabled me to accomplish the tasks. *

Mark only one oval.

1 2 3 4 5

Fully ☐ ☐ ☐ ☐ ☐ Fully agree

22. I would use the search engine again. *

Mark only one oval.

1 2 3 4 5

Fully ☐ ☐ ☐ ☐ ☐ Fully agree

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